

Ashvin Singh

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SUMMARY

Full-stack developer and B.Tech Computer Science (AIML) student with hands-on experience building AI agent systems, REST APIs, and production web applications. Shipped multi-agent orchestration engines, MCP-based memory systems, and OAuth-integrated full-stack apps using React, FastAPI/Flask, and SQL databases. Comfortable across the stack – frontend, backend, and database design – and quick to pick up new frameworks in fast-moving, agile teams.

EDUCATION

VIT Bhopal University

B.Tech in Computer Science & Engineering (AIML) – GPA: 8.18/10.0

- Relevant Coursework: Data Structures & Algorithms, DBMS, Machine Learning, Deep Learning, Artificial Intelligence, Data Science

Bhopal, MP, India

Expected Aug. 2027

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, C++, C

Backend & APIs: FastAPI, Flask, REST API Design, SQLAlchemy, SQLite, PostgreSQL (familiar), OAuth 2.0, Gmail API, Google Tasks API

Frontend: React, Vite, HTML, CSS

AI / Agents: Multi-Agent Orchestration, MCP (Model Context Protocol), LLM Integration, NLP, PyTorch, TensorFlow

Tools & Practices: Git, GitHub, Linux, Agile Development, CI/CD Basics, OOP, DBMS

PROJECTS

Inbotic | FastAPI, SQLAlchemy, React, Gmail API, Google Tasks API, LLM, SQLite

Jan. 2026

- Built and deployed a full-stack multi-user web app that automatically extracts dates from Gmail messages and converts them into Google Tasks using LLM-assisted parsing, live at inbotic.onrender.com
- Engineered a FastAPI backend across 8 modules with Google OAuth 2.0 authentication, rate limiting, and dual interface layers (Jinja2 server-rendered and React/Vite SPA)
- Integrated 3 Google APIs end-to-end (Gmail, Tasks, OAuth) with SQLite-backed persistence for user, task, and email state

BaseMem | Python, Flask, SQLite, MCP, D3.js, tree-sitter

Apr. 2026

- Built a persistent memory system for AI coding agents, exposing 30 MCP tools across a CLI, MCP server, and Flask REST API, integrated with 5+ agent platforms including Claude Code, Cursor, and Windsurf
- Designed a weighted knowledge graph on SQLite + FTS5 with Jaccard-similarity auto-linking, edge reinforcement, and 3-tier memory management (hot/warm/compacted)
- Implemented tree-sitter-based code intelligence indexing across 300+ languages

MAOE – Multi-Agent Orchestration Engine | Python, SQLite

2025

- Architected a multi-agent system using Blackboard Architecture with 5 distinct agent roles coordinating through a shared, thread-safe blackboard with auto-compression and TF-IDF-based deduplication
- Added SQLite persistence so agent state and task history survive across sessions

PROFESSIONAL EXPERIENCE

Core Member

College Tech Club, VIT Bhopal

Mar. 2024 – Mar. 2025

Bhopal, India

- Coordinated a university-wide hackathon across 3 judging rounds with 50+ participants; crafted Figma prototypes and improved average submission responsiveness by 30%
- Led outreach campaigns across 4+ campus platforms, reaching 1,000+ students and increasing event participation

EXTRACURRICULAR ACTIVITIES

Open Source Contributor

Python AI & Web Framework Projects

2023 – Present

GitHub

- Submitted 5+ contributions (bug fixes, documentation, features) to Python-based open-source projects; earned the Pull Shark GitHub Achievement
- Actively participated in hackathons, coding competitions, and data science challenges on Kaggle, LeetCode, and Codeforces

CERTIFICATIONS

Machine Learning Specialization – Andrew Ng, Coursera | C++ Programming Certificate – Udemy